



Simon A. Babayan
Institute of Biodiversity, Animal Health,
& Comparative Medicine
Graham Kerr Building
Glasgow, G12 8QQ
Email: simon.babayan@glasgow.ac.uk

October 31, 2022

Editorial Board of *eLife*.

Dear editors,

We are pleased to submit our final revision of the manuscript entitled “**Causes of variation in vaccine efficacy in a wild rodent population**” for publication in PLoS Biology.

In this paper we address both the fundamental methodological issues of transportability, i.e., generalising laboratory-based results to their intended target population, and we explore the mechanistic causes for the loss of vaccine efficacy that are typically overlooked in preclinical phases of vaccinology. To do this, we used a well-characterised rodent species, *Apodemus sylvaticus*, which is native to Europe and that has a characteristically rich parasite community (unlike *Mus musculus*). The power of this system is that it allows experimental manipulation as well as longitudinal and cross-sectional sampling in both the laboratory and natural populations. In this way, it is unique in its ability to allow disentangling complex processes that other systems might capture but be unable to resolve.

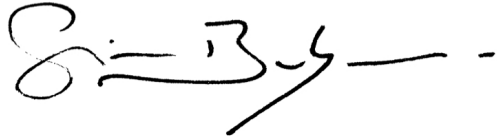
The main novelties of this paper include:

- Direct comparison of vaccination in laboratory and wild habitats using the same model species, vaccine regimen, and sampling
- Application of structural causal modelling to the identification of the main drivers of variation in vaccine efficacy across laboratory and wild habitats
- Elucidation of the relative contribution of habitat, diet, and demographic factors to gains or losses in vaccine efficacy.
- Generation of novel insight and hypotheses that might explain some discrepancies between prior reports of the effects of diet and parasite infection on vaccine immunogenicity

We therefore expect this study to be of broad appeal beyond the specific biological system in which it was conducted: we bring further support to the idea that the immune system adopts alternative strategies depending on diet quality; that parasite infection interacts with diet in regulating adaptive immunity; that males consistently generate lower antibody responses to immunisation.

This manuscript was published on BioRxiv at <https://add.url>. In addition to the files submitted for this initial submission, we provide the Overleaf (using the eLife LaTeX template) link to the manuscript: <https://www.overleaf.com/read/>

Sincerely, on behalf of my co-authors,

A handwritten signature in black ink, appearing to read 'S. Babayan', followed by a horizontal line and a dash.

Simon A. Babayan, PhD